Chapter7: p Group

Saturday, 23 March 2024

Defn 7.1 Conjugation :- the operation of G on itself is defined by

$$(g, x) \rightsquigarrow gxg^{-1}$$

Defn 7.2 Center of a group G is

$$Z = \{ x \in G \text{ s.t } x.g = g.x \ \forall g \in G \}$$

Defn 7.3 Centralizer of an element x of an group G

$$Z(x) = \{ g \in G \text{ s.t } gx = xg \}$$

Lemma 7.4 $Z(x)$ contains → x, Z, Maybe some other elements

Lemma 7.5

$$|G| = |Z(x)| \ |C(x)|$$

Conjugacy class of x

Lemma 7.6

$$|G| = |C_1| + |C_2| + \dots + |C_k|$$

Lemma 7.7 If $x \in Z$ then it's conjugacy class contains only x itself.

Defn 7.8 p Group :- A group is called p group if its order is positive power of a prime number

Lemma 7.9 The center of a p group is non trivial

Lemma 7.10 Let G, a p group, acts on a finite set S of order n not divisible by p. Then S contains atleast one element which is fixed by all elements of G

***Lemma 7.11** Every group of order p^2 is abelian

*Lemma 7.11 Every group of order p^2 is abelian

* Lemma 7.12 A group of order p^2 is either cyclic or isomorphic to the product of two cyclic groups of order p

Chapter7: Conjugation in Sn

Saturday, 23 March 2024

* Defn 7.13 A group G is called simple if it has no proper normal groups.

* Lemma 7.14 If a normal group N contains an elements x, then N also contains conjugacy class of x.

* Lemma 7.15 $p, p' \in S_n$ are conjugates if and only if their cycle decompositions have the same order.

* Lemma 7.16 For $n \geq 3$ A_n is generated by 3 cycles

* Lemma 7.17 For $n \geq 5$, the 3 cycles form a single conjugacy class in A_n

* Theorem 7.18 For $n \geq 5$, $A_n \subset S_n$ is a simple group

* Defn 7.19 Let $H \subset G$. Consider conjugation

Orbit of H = conjugate subgroups of form gHg^{-1} where $g \in G$

Stabilizer of H = $N(H) = \{ g \in G \text{ s.t } gHg^{-1} = H \}$

* Lemma 7.20 $\rightarrow H \triangleleft N \subset G$

$\rightarrow H \triangleleft G$ if $N(H) = G$

* Lemma 7.21 If $|S| = n = p^e m$, where m is not divisible by p. Then

Let $U = \{$ subset of S of order $p^e \}$

$|U| = N = {}^{n}C_{p^e}$ is not divisible by p

* Theorem 7.22 If G is a group, U is its subset and H is stabilizer of U. Then $|H|$ divides $|G|$ and also $|H|$ divides $|U|$

Hint: $H \times U \longrightarrow U$

Right cosets [$\because$ same size as $|H|$]

$$|U| = |O_{u_1}| + |O_{u_2}| \; __ + |O_{u_k}|$$

Chapter 7: Sylow's p subgroup

Saturday, 23 March 2024

* Defn 7.23 — If $|G| = p^e m$ where $e \geq 1$ and p doesn't divide m. A sylow p subgroup is the subgroup of G of order p^e

* First Theorem 7.24 — If $|G| = p^e m$, where $e \geq 1$ and p doesn't divide m. Then there exists sylow's p subgroups of G.

Hint $G \times S \longrightarrow S$, S: set of subsets of G of order p^e

* Second Theorem 7.25
- If G contains an p subgroup K. Then $K \subset gHg^{-1}$, where H is an sylow's p subgroup
- All the sylow's p subgroups are conjugates of each other

Hint: $G \times G/H \longrightarrow G/H$ (left multiplication, transitive) H: Sylow p group

$c = [H] \in G/H$ with stabilizer H

$K \times G/H \longrightarrow G/H$ K: p group

$|G/H|$ not divisible by $|K|$

$\therefore \exists c' \in G/H$ stabilized by K

Also since $G \times G/H \longrightarrow G/H$ is transitive

$c' = gc$ and $H_{c'} = gHg^{-1}$

$K \subset H_{c'} = gHg^{-1}$

* Third Theorem 7.26 — If s is the number of sylow's p subgroups. Then
1. s divides m
2. $s = kp+1$ where $k \in \mathbb{Z}$

Hint $G \times S \longrightarrow S$ (conjugation, transitive) S: Set of Sylow p subgroup

$[H] \in S$ Sylow p subgroup

$|G| = |N(H)|\,|O_H|$

$= |N(H)|\,|S|$

$\therefore$ s divides m

$H \subset N(H)$

$\therefore p^e \leq |N(H)| \leq p^e m$

$H \times S \longrightarrow S$ (conjugation)

.. s divides m

conjugation
$H \times S \longrightarrow S$

show only $[1] \in S$ has orbit of order 1

Chapter 8: Hermitian Form

Saturday, 23 March 2024

Normal Matrix
$A^*A = AA^*$

* Defn 3.1 Symmetric Matrix
$A^T = A$

Orthogonal Matrix
$A^TA = I$

Hermitian Matrix
$A^* = A$
conjugate transpose

Unitary Matrix
$P^*P = I$

* Defn 3.2 Symmetric Bilinear Form
$\langle , \rangle : V \times V \longrightarrow \mathbb{R}$ s.t
1. Linear in first variable
2. " " second "
3. $\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V$

Hermitian form
$\langle , \rangle : V \times V \longrightarrow \mathbb{C}$
1. Conjugate linear in 1st variable
2. Linear in 2nd variable
3. $\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \forall v, w \in V$

* Defn 3.3
Matrix of form : A where
$a_{ij} = \langle v_i, v_j \rangle$
$[B] = (v_1, v_2, \ldots, v_n)$
basis

* Lemma 3.4 Matrix of the form is Symmetric

Matrix of the form is Hermitian

* Lemma 3.5 $\langle v, w \rangle = X^T A Y$

$\langle v, w \rangle = X^* A Y$

where X and Y are coordinate vectors of v, w w.r.t $[B]$

* Lemma 3.6 If $[B]$ and $[B']$ are two basis of V s.t $[B'] = [B]P$ and A, A' are matrices of $\langle , \rangle$ wrt to $[B]$ and $[B']$ then

$A' = P^TAP$

$A' = P^*AP$

Chapter 8: Orthogonality

Saturday, 23 March 2024

* Lemma 8.1



$\forall v, w \in V$ and $v, w \neq 0_V$

* Defn 8.8 $v \perp w$ if $\langle v, w \rangle = 0$

* Defn 8.9 Orthogonal space $W^\perp$ to subspace $W \subset V$

$$W^\perp = [v \in V \text{ s.t } \langle v, w \rangle = 0_V \ \forall w \in V]$$

* Defn 8.10 $v \in V$ is called a null vector if

$$\langle v, w \rangle = 0 \quad \forall \ w \in V$$

* Defn 8.11 Nullspace N of $\langle\ \rangle$ is the subspace of null vectors.

$$N = [v \mid v \perp V] = V^\perp$$

* Defn 8.12 $\langle , \rangle$ is called non degenerate on V if $N = \{0_V\}$

* Defn 8.13 $\langle , \rangle$ is called non degenerate on $W \subset V$ if $W \cap W^\perp = \{0_V\}$

another way of saying the same thing

if $w \in W$ and $w \neq 0_V$ then $\exists\ w' \in W$ s.t $\langle w, w' \rangle \neq 0$

* Caution A form may be non degenerate on the whole vectorspace and degenerate on subspace and vice versa.

* Lemma 8.14 $\langle , \rangle$ non degenerate $\iff$ matrix of the form A is invertible.

Chapter 8: Orthogonal basis

Saturday, 23 March 2024

* **Theorem 8.15** $\langle , \rangle$ is non degenerate on $W \subset V$ if and only if $V = W \oplus W^{\perp}$

* **Lemma 8.16** If $\langle , \rangle$ not identically zero $\exists\, v \in V$ s.t $\langle v, v \rangle \neq 0$

* **Theorem 8.17** Given $\langle , \rangle$ on V, there exists a orthogonal basis of V wrt $\langle , \rangle$

* **Lemma 8.18** Given $\langle , \rangle$ on V, there exists a basis of V s.t the matrix of the form has this structure

$$A = \begin{bmatrix} I_m & & \\ & -I_n & \\ & & 0 \end{bmatrix}$$

* **Defn 8.19** If $\langle , \rangle$ is non degenerate on $W \subset V$ then $V = W \oplus W^{\perp}$

Orthogonal Projection (linear transformation) $\pi : V \longrightarrow V$ s.t

$\pi(w) = w \quad \forall\, w \in W$

$\pi(w') = 0 \quad \forall\, w' \in W^{\perp}$

* **Lemma 8.20** Projection Formula : If $[v_1, v_2 \cdots v_k]$ is orthogonal basis of W then

$$\pi(v) = c_1 v_1 + c_2 v_2 \cdots + c_k v_k$$

where

$$c_i = \frac{\langle v, w_i \rangle}{\langle w_i, w_i \rangle}$$

* **Defn 8.21** Euclidean Space: Real Vectorspace V + ve definite symmetric bilinear form

* **Defn** Hermitian Space: Complex Vectorspace V

* Def^n 8.22 Hermitian Space: Complex Vector space V + +ve definite Hermitian form

Chapter 8: Normal Operator

Saturday, 23 March 2024

* **Lemma 8.23** If V is a Euclidean or Hermitian space. If $W \subset V$ then the form $\langle , \rangle$ is non degenerate on W and $\therefore$ $V = W \oplus W^{\perp}$

* **Defn 8.24** $T: V \to V$ be a linear operator on a Hermitian space.
  T is called
  normal if its matrix A wrt a orthonormal basis has : $A^* A = A A^*$
  hermitian " " " A " " " " : $A^* = A$
  Unitary " " " A " " " " : $A^* A = I$

* **Lemma 8.25** If A is a normal, hermitian or unitary then $P^* A P$ is also the same, where P is any unitary matrix.

* **Defn 8.26** If $T: V \to V$ is a linear operator on Hermitian space.
  Adjoint operator $T^*: V \to V$ where the matrix of T^* is A^*

* **Lemma 8.27** Let T be a linear operator on Hermitian space and T^* be the adjoint operator.

  $\forall\, v, w \in V$

  (a) $\langle Tv, w \rangle = \langle v, T^* w \rangle$ and $\langle v, Tw \rangle = \langle T^* v, w \rangle$
  (b) if T is normal $\langle Tv, Tw \rangle = \langle T^* v, T^* w \rangle$
  (c) if T is Hermitian $\langle Tv, w \rangle = \langle v, Tw \rangle$
  (d) if T is Unitary $\langle Tv, Tw \rangle = \langle v, w \rangle$

* **Lemma 8.28** Let T be a linear operator on Hermitian space V. Let $W \subset V$
  W is T invariant $\iff$ $W^{\perp}$ is T^* invariant

* Theorem 8.29 If $T: V \to V$ normal operator on a Hermitian Space V.
If (v, λ) is eigen vector of $T \Rightarrow (v, \bar{\lambda})$ is eigenvector of T^*

Chapter 8: Spectral Theorem

Saturday, 23 March 2024

* **Spectral Theorem 8.30**: If $T: V \to V$ is a normal operator on a Hermitian space. There is an orthonormal basis of V with eigen vectors of T.

* **Lemma 8.31**: If A is a normal matrix. There is unitary matrix P s.t P^*AP is diagonal

* **Corollary 8.32**: Every conjugacy class of a normal matrix contains a diagonal matrix

* **Corollary 8.33**: If $T: V \to V$ is an Hermitian operator on a Hermitian space. Then the eigenvalues of T are real.

* **Theorem 8.34**: If $T: V \to V$ is an symmetric operator on a Euclidean space. Then there is a orthonormal basis of V consisting of eigen vectors of T.

* **Lemma 8.35**: If A is a symmetric matrix then $\exists$ a orthonormal matrix P s.t P^TAP is diagonal

* **Defn 8.36**: A bilinear form $\langle , \rangle$ on a vectorspace is called skew-symmetric if

$$\langle v, v \rangle = 0 \quad \forall v \in V \qquad \text{—①}$$

or

$$\langle u, v \rangle = -\langle v, u \rangle \quad \forall u, v \in V \qquad \text{—②}$$

These conditions are same if the characteristic of F is not 2.
If it is 2 condition ① is correct

* **Lemma 8.37**: If $\langle , \rangle$ is a non degenerate skew symmetric form on vector space V the $\exists$ a basis $[B]$ of V s.t the matrix of the form has the following structure.

$$\begin{bmatrix} & \end{bmatrix} \qquad \begin{bmatrix} & \end{bmatrix}$$

2.57 the $\exists$ a basis $[\mathcal{B}]$ of V s.t the matrix of the form has the following structure.

$$S_0 = \begin{bmatrix} \varepsilon & & \\ & \varepsilon & \\ & & \ddots \end{bmatrix} \quad \text{where } \varepsilon = \begin{bmatrix} -1 & \\ & 1 \end{bmatrix}$$